Lesson plan · for the teacher

Build a rain gauge

IoT · Level 1 · Lesson 3

Time
40 min

Materials
One clear plastic bottle with straight, smooth sides per group; Scissors or a knife for the teacher only; tape; Small stones, water, a ruler and a marker or pen; Paper strips for a millimetre scale

Learning goals

- Explain that rainfall is measured in millimetres of depth.
- Build a rain gauge from a bottle, with a zero line, a millimetre scale and a funnel.
- Choose a good place for the gauge and agree a daily reading routine.

Activity: Make a bottle rain gauge: Build

1. Warm-up (5 min): Ask: “Did it rain a lot or a little last time?” Let pupils disagree. Write their words on the board — “a lot”, “a bit”, “heavy”.
2. Teach (7 min): Explain millimetres of rain: the depth on flat ground. Show a wide and a narrow straight-sided container and ask: in the same rain, which fills deeper? Explain that both reach the same depth.
3. Build (15 min): Groups add stones and water just to cover them, mark the 0 line, and tape on a paper scale marked every millimetre from a ruler. Then they turn the cut top into the bottle as a funnel and tape it.
4. Place it (8 min): Walk to the spots you chose. Groups judge each one: open sky? Far from roofs and trees? Safe from feet and animals? Place the gauge in the best spot.
5. Wrap-up (5 min): Back in class, draw a reading rota and a table on the board: date, time, rain (mm). Exit ticket: one rule for reading the gauge.

Before class, cut the top off each bottle where the sides start to curve, and tape the cut edges. Check that each bottle’s sides are straight. Walk the school grounds and pick two or three open spots to discuss.

Discussion

- Is your 0 line level?
- Does every millimetre on your scale match the ruler?
- Is there anything above the gauge that could block rain or drip water in?

Exit ticket

- One rule for reading the rain gauge, and why it matters.

Try it in class tomorrow

Build a class rain gauge: Bring a clear, straight-sided bottle. Cut off the top yourself, tape the edge, and let pupils add stones, water to a 0 line, a paper mm scale and the funnel. Agree an open spot and a daily reading time, and draw a rota on the board.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Your school has three possible spots for the rain gauge. Which is best?**

- ◯ In the middle of the open school field
- ◯ Under the big tree by the gate, out of the wind
- ◯ Beside the classroom wall, under the roof edge

2 **Put the steps for building and using the class rain gauge in order.**

- ☐ Write it down, then empty it back to 0
- ☐ Read it at the same time each day
- ☐ Set it in open ground, away from trees
- ☐ Add a mm scale and the funnel
- ☐ Add stones and water; mark the 0 line
- ☐ An adult cuts off the bottle's top

3 **A wider bottle will show more millimetres of rain than a narrow one.**

◯ True ◯ False

4 **Match each part or habit to its reason.**

Straight sides	• • • •	No rain is counted twice
Upside-down top as a funnel	• • • •	Each reading covers one full day
Stones in the bottom	• • • •	Stop it blowing over
Same time each day	• • • •	Guides rain in, keeps most leaves out
Emptying after reading	• • • •	Every millimetre on the scale is true

5 **Complete the sentence.**

Rainfall is measured in _____ : the _____ the rain would reach on flat ground.

Word bank: millimetres · depth · litres · weight

Where in your school could a rain gauge stand in the open, safe from animals and balls? Who would read it each day?